

# Active Inference Institute ~ 2023 Quarterly Roundtable #3

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Hello everyone, it's September 29th, 2023. We're here in the Active Inference Institute's quarterly roundtable number three. So this is our third of the quarterly updates. We have guests. I'm Daniel, one of the officers of the Institute, and I'm Blue Knight, also another officer of the Institute. All right, we are a participatory online institute that is communicating, learning, and practicing applied active inference. You can find us at [activeinference.institute](https://activeinference.institute) or the [.org](https://activeinference.org) address. We'll be moving things over more to the [.institute](https://activeinference.institute) address though. We're inference active on Twitter or X. And here's more information. This is recorded in an archived live stream. So please provide us with feedback so we can improve on all aspects of our work. All backgrounds and perspectives are welcome, and we'll follow video etiquette for live streams. So today in this quarterly update, we will first cover some big developments and updates at the institute and ecosystem scale. Then go over some educational and research updates from our Eduactive and re-inference organizational units respectively. And then have some fun and entertaining musings on where we're going for the last months of 2023, where we can all go in 2024. And if you're watching live, please feel free to add any questions in the live chat. And then in the last section, we'll be able to scan over that and look at your questions. All right, first updates from the institute at the organizational scale. One of the big highlights of the last months and really of the year, and of really all of our work together was the accomplishment of this collaborative paper that we released in August of this year, the Active Inference Institute and Active Inference Ecosystem. We had a great team of co-authors. And to give a visual abstract and just a teaser for those who might want to learn more about this paper, we went through a great collaborative process to document and align on various aspects of what had been, was, and will be happening. And on the right side is figure one. This is an overview of the institute's structure itself in engagement with the ecosystem. On the left side are a representation of our open source products and some of the functions that they provide in the ecosystem. There's a lot more to say, but Blue, what do you want to comment on? Or what was it like for you?

It was super intense. I think that we applied for a phase two grant, which I think we should talk about maybe. And in doing that phase two pose grant, this was the preparation that they kind of give you in the phase one. And it really is about becoming a big business or a big corporation or a big organization. Right. And like what it takes to really get organized and anchor and manage a large community of people, which I think that was really good for us. So like we are already kind of, I don't know how much managing we do.

We're a very distributed system, but we do already oversee or engage with a large community of people. And like, what does that look like when maybe we have conflicting factions or things get dropped, like the ball gets dropped in one court or another? So, so I think that that really kind of brought to light a lot of the things we need to work on and also the things that we're doing a good job at.

Awesome. Yeah. We'll come to the grant in a few slides, but just to verbally describe the image on the right, we have the circle, a golden ring that alludes to the cover of the Par, Pizzullo and Friston 2022 textbook. And that ring symbolizes the active inference ecosystem and also the boundary between the institute as an entity

and the ecosystem as niche. In the institute itself, starting on the top of within the ring, we have several official bodies or organs. We have the board of directors who select officers, as well as scientific advisory board, which is an informal guidance and insight group, which we'll come to later. And the institute provides institute level functionality and support for the two organizational units, which are Eduactive, which is focused on educational outcomes and re-inference, which is focused on research work. The organizational units host through facilitation, other processes, different projects that are all sorts of within, on, and out there in the ecosystem, and then out in the ecosystem, we have all different types of activities and products and organizations and players.

And if you're listening to this, we encourage you to get involved. If it's something that you want to learn more about,

then the paper is a great place to start. Best starting place that we've made so far.

And the next step is to take that active step and actually get involved. Anything else to say on this?

Just, we should probably highlight the fact that our research unit, which hasn't been formally, like, we've formally renamed our research unit to re-inference, which we haven't announced yet.

So I think like, as we are talking about the re-inference unit, we did that kind of in the construction of this paper. And because we were, we've been Eduactive and research, and we just decided that re-inference was more fun.

Kind of, you know, shouting out to active inference, the whole thing, not just the active part.

Fun. Yeah. So thanks again to all the authors here and also people who contributed their insights and just didn't want to be co-authors.

All right. Carrying on with Institute level updates. On August 22nd of this year, we had a really amazing applied active inference symposium.

This was our third such symposium. The first was in 2021 with Carl Friston as a kind of one person symposium.

Last year in the second symposium, we highlighted robotics and a variety of theoretical applied work in robotics.

This year we built off of the December 2022 framing of Friston et al designing ecosystems of shared intelligence, and we wanted to put it into play.

And so we co-organized with a really great team enacting ecosystems of shared intelligence.

And here are the presenters and discussants who joined and enacted the symposium.

And there's two intervals. It was an awesome symposium. Super exciting to hear the breadth and the depth of the work happening in the ecosystem.

So for here, all I'll say is check out the program and recordings and transcripts if you want to learn more about that.

It was an epic symposium, though. And thanks again to the co-organizer team.

Anything you want to add, Blue?

Okay.

Okay.

As Blue alluded to earlier, we applied for a phase two grant within the NSF program pathways to enable open source ecosystems.

So here's just two visualizations that are provided by the NSF to describe a little bit of what they mean by an open source ecosystem.

On the right side, there's an open source product.

And then through a pathway to sustainability, an open source ecosystem can exist, which includes here not just scientist, open source product, and external user,

but governance, documents, scientists, onboarding support, external users, the product, developer, community, distributed, defect management, development, and integration and security.

And it was super fun to think about our work in this broader sense of open source products.

Open source doesn't only refer to code.

So thinking about what does a really rich, multimodal open source product look like and what is it?

And you want to add anything there, Blue, or I can add more.

Yeah.

I just said our...

So the paper that we wrote was kind of the scoping document for the grant, and it's what the people do in phase one.

And in terms of open source, I think, like, we have done everything open source at the Institute.

Like, all of the things that we've done from the beginning of, like, our formulation until now.

So it was kind of good to have that acknowledged that, like, you can have an ecosystem around education or around, like, just live stream videos or information sharing, information management.

Like, we've really built up the active inference community, I think.

I think we've been a value add.

Because somebody has to kind of organize and give meaning to kind of the random contributions.

Or not random, but I think, like, it's helpful in a field that's so new and then moves so quickly in terms of research developments and everything else.

Yeah.

Thanks.

Thanks.

Just to give a little more detail, the primary submitter on this grant was the University of Washington Applied Physics Lab via co-PI Scott David at the IRS IRI, as well as Blue and I.

And as Blue mentioned, it was really fun to see who was ready for what, who wanted to throw a value into the arena, who wanted to put a comment on a Google Doc, who is ready to state their hourly rate and scope of work and get involved if there's pragmatic value of that kind on the line, who's ready to work in the meanwhile.

And so it was just many discussions and negotiations.

And as we can smile and laugh at now, a lot of curveballs and things happening.

So it was great times.

Okay, moving on.

Just within the last weeks, we have submitted our Form 1023 to the IRS.

And so we soon expect and prefer ourselves to be a 501c3 tax-exempt organization, which means that donations that people provide to us, of which we have received none so far, will be tax-exempt.

Once that status and all associated infrastructure is ready, we'll begin philanthropic campaigning.

With Blue especially, we are working on policies and procedures proactively so that we can have compliance ahead of time and just stay right and relevant.

It would be extremely helpful and much appreciated if anybody was interested in contributing their experience, networks, or resources to what it could be for a philanthropy or fundraising program at the Institute.

We're open to how that might develop.

And so if you see a high leverage opportunity for what you know or who you know or however, then some truly amazing things could happen.

And if you know, you know, and if you know, then help.

Anything you want to add on this?

I do want to add that it was really interesting in developing ecosystem infrastructure in the POSE grant and in the ecosystem paper.

It was really interesting to see what it takes to be a sustainable ecosystem, to really evaluate that at the level of the Institute and the ecosystem.

And I do think that, I mean, what makes an organization dissolve?

Like people either like, you know, it's like time and people and money, right?

It's like time and money, time and money.

As long as people have time and people have money and interest level.

So we have a great like interest level as we'll talk about, I think, in the coming few slides.

We'll demonstrate that.

But how do we get people to keep having time to commit to the Institute and the organization?

So I think that one of the key things that we need to do is fundraise.

So like philanthropy also, but also grant writing.

So these two things will be, I think, central to keeping the Institute, like being the glue that keeps the Institute together in the coming years.

So I just want to just highlight that for anybody who wants to come be some glue, come provide some adhesion in the niche.

That'd be great.

Yeah, it kind of goes without saying, but it's easy to skip.

Everything that has happened so far has been a purely volunteer effort, including this very little live stream.

And the last few months have signaled and anticipated and readied for literally the first paid people to do

anything for the Institute, which is a big game changer and would scaffold our time.

And so that will be quite a transition point.

Let's continue.

Also at the Institute level, we continue with our volunteer program.

As always, I can do nothing but encourage and exhort you to volunteer and get involved at the Institute.

Being a volunteer supports your engagement with projects and learning groups, whichever ones you're interested.

It gives you repeatedly affordances provided to get involved with Active Inference and your contributions will matter.

The questions you ask on a live stream will be published by the Active Inference Journal.

The questions that you write in the textbook group, CODA, those do become part of a shared knowledge niche.

Code developments that you provide, something that makes it more accessible or applicable, those are huge in the setting that we are and will be in.

So please volunteer and encourage other people who could be looking for technical or philosophical education, community, mentorship, all these different things.

This is a great way.

And for those who would like a little bit more structured of an approach to getting involved, we continue to have the internship program.

There's about 20 active interns from around the world who are engaging with different research and education affordances.

And we are up to now five successfully completed internships.

So to those who completed, it's been honor and an epic time.

There's so much more I could say, but it's really exciting what interns come in with because we ourselves or from what we hear,

people are excited to apply active inference and to improve their own education and understanding of what active inference is and how they might use it or grasp or handle it.

And then to have this lightweight internship program that makes a connection between the institute and the person,

gives a container and a name and a timeline and an asynchronous document for people's engagement, whether they're in a traditional educational program, like an undergraduate student or a graduate student, or probably the majority of which are outside of a traditional educational program and just lifelong learners and lifelong cool people.

And they just come to work and hone what they do.

So it continues to be very inspiring and I encourage people to be interns.

And also for those who would like to facilitate or mentor,

we would like to start that discussion about how that might work.

Anything else you want to add?

The Active Inference Matchmaking Service.

There's so many funny ways to take that, but yeah.

What do you expect and prefer?

I bet like matching mentors and interns.

Yeah.

That's it.

Of course.

And then just a little summary update across our different platforms and public interfaces.

We see continued growth and activity.

Just one plot.

This is our YouTube channel growth through time.

Just kind of interesting.

Slow and steady wins the race, but it's also always good to see that first and potentially second and third order derivatives increasing.

Anything else you want to add?

No, I think like the YouTube channel has really blown up like in the last quarter.

And I think a lot of that, we just have to shout out to the Chris Fields course because the views are astronomical in those videos.

It's like really, it's really blowing up.

So thanks, Chris, if you're watching and to everybody who's viewing those videos and taking those courses, which it's truly awesome, actually.

Yeah.

Yeah.

We'll return to that, but good point.

And also for people who are interested in even social media management, video production, podcast editing, discord moderation and governance, meta governance.

There's a lot of things that don't require some technical fluidity with the active inference kernel.

So there's a lot of ways that people can help.

So don't let it provide too much friction or freeze you out if you feel like you want to be involved, but you wonder if you don't know enough about some technical area or philosophical framing.

It's totally not required.

And there's a million ways that we can work with wherever people are.

And actually, that's like, just like, that's like one of the key areas that we've identified in our ecosystem oversight.

It is social media management.

So it's not, you know, all about the math.

Like, so, so many people come to us like, I'm so curious, except I can't do math.

And it's okay, actually, like, can you do Twitter?

Because really, there's a lot of opportunities for involvement with the Institute that aren't math heavy.

Before we head into the organizational units, here's the big picture on which activities are happening.

This website, which you'll find heartily linked from basically anywhere you are, always summarizes the up-to-date representation of what activities are happening and what you can do to get involved.

So I'll just go over this and we'll talk about a few of these projects in the coming slides.

So in the space of publishing different medias, we have the Livestream Project, where we organize and prepare for different Livestreams.

Following Livestreams, transcripts are created and published through the Active Inference Journal, which, through the work of Alex, Holly, Dave, others, is always improving.

And it's an exciting time for all of these natural language processing analyses as well.

This is all being pushed to the GitHub that allows you to clone off the current state of the art in all the transcripts and translations.

In the learning groups area, we have the PARDAL textbook group, cohorts 4 and 5, continuing through the textbook group,

Fundamentals of Active Inference, Sanjeev Namjoshi's book, In Progress,

and two courses which we'll come to later, Physics is Information Processing by Chris Fields and Constructing Cultural Landscapes, Active Inference for the Social Sciences by Avell et al.

In the software space and activity, we continue on Active Lockference, where we work from both ends to make cognitive modeling more accessible and applicable.

We work from the technical capacity side to develop the Active Blockference project itself as a package, utilizing CAD-CAD to apply ACTIMF generative models.

And we also work from the system or domain of interest side, working with people wherever they're at in their analytical or programming abilities and just starting to get the inklings of the recipe for generative modeling together.

Robotics and Embodied, JF continues to develop his work with LEGO robots and societies of mind.

Active Inference Ontology also continues maintaining and slightly improving, as well as extending the Active Inference Ontology.

And we have two weekly discussion sessions, one more on the educational theme, one more on the research theme, but ultimately they're both just how it goes in the moment.

Anything to add?

All right.

In Eduactive, our educational organizational unit, one of the most exciting continued activities is the Active Inference Textbook Group.

Cohort 3, just within the last weeks, finished their journey.

So congrats to those who stuck it through Cohort 3.

Cohort 4 is now in Chapter 6, Recipe for Generative Modeling, continuing on with the book.

And Cohort 5 is starting off there in Chapter 1.

Everyone's welcome to join either or both of these cohorts,

which are staggered so that you can attend the same time zone and get sampling across the book or you can stay on one trail at a time and switch time zones. Everything's recorded.

Everything is collaborative in our shared workspace.

It's always possible to register and get involved.

All backgrounds are totally welcome and encouraged.

So if you've seen this textbook and or you want to learn more about what the only Active Inference Textbook in the play right now says,

and also connect a ton of dots and bring in a ton of other resources and more, then the Active Inference Textbook Group is a great place.

And you'll meet many people who are also interested in similar topics.

Anything else, Blue?

So it's fun.

Okay.

The course that began in May 2023 is now about five-sixths completed.

That's Physics as Information Processing by Chris Fields.

It's been a journey.

And can't wait for the fifth discussion section, which is tomorrow, and then the sixth lecture and discussion.

And that will have locked in the completion of the first course at the Institute.

We'll be able to publish the proceedings of the lectures and discussions, as well as the asynchronous question and answer, which Chris Fields is very graciously answering.

Then we'll be able to explore cohort or individual-based learning structures around this really first principles and groundbreaking content that Chris is sharing.

Want to add anything on this one?

Just wow.

That's all.

Yeah.

Also in the courses space, we continue on with the Constructing Cultural Landscapes, Active Inference for the Social Sciences course, with Avel, Ben, Mao, Lorena, and myself.

So Ben, Mao, myself, and Lorena have completed their sections with a lecture and a discussion, although I guess Mao's discussion is coming up.

We look forward to also the continued activity in this course.

As we remarked on in our last quarterly roundtable, second one,



this year has and will see us go from zero courses to two completed courses.

So both the physics and the social sciences,

which under some understandings run the gamut of multiscale systems,

to have both of those at such high quality,

volunteered and catalyzed by real unique expertise

and also broad and diverse participation is incredible.

Anything you want to add, Blue?

Just that as we are starting to have courses hosted by the Institute

and completed,

anybody interested in having a course hosted by us should reach out.

And anyone familiar with or interested in accreditation

is also something like we've been thinking about.

How do you, you know,

what would certification in Active Inference look like?

What are the key buildup skills?

And how could we develop a program to say someone is certified in Active Inference?

It's something that we've been thinking about at the Institute scale.

So if you have ideas, we would love to hear about them.

And if you have experience, we would love to get your input.

Yeah.

It's not covered in this roundtable,

but the qualification certification,

something that we're thinking about

and the pedagogy and developing and hosting courses,

these are all things that we will work on.

In the education space of live streams,

we continue to have exciting and unique live streams.

Here's just kind of a slice of life.

Here's the upcoming live streams for the coming 30 days.

We have the roundtable that we're in right now,

strangely looping.

We have Chris Fields' discussion tomorrow.

We have social sciences discussion next week.

Lecture 6 and discussion 6 for physics.

Active inference for the social sciences.

Avel's lecture and discussion.

And a guest stream on conviction narrative theory.

And a new live stream paper-based series

kicking off on message passing.

So it's just kind of like a flavor of sometimes the work to prepare for this live stream is just to coordinate with a guest.

And it doesn't require any kind of expertise in their topic.

Actually, just a curiosity in it.

Other times, we are operationally or logistically supporting the course teachers and also just showing up.

And then there's also work like the .zero preparation,

where for weeks to months ahead of the live stream,

we're studying through a paper

and leveraging the questions that we have

by asking early and asking often.

And precisely, or at least legibly,

that enables the questions that are asked before the live stream

by the .zero preparers

to then actually realize the addressing of those questions

instead of a thousand scattered notebooks after the fact,

or there being no fact in just a thousand scattered notebooks,

the .zero, I'm very passionate about

because it is a way to do the homework early

and by doing so, change how things actually happen.

So if you're interested in having any kind of impact

or learning with the live stream organizing and preparation process,

whether you have this or that familiarity with the video

and the media and the communication side,

or with any topics in and of themselves,

it's super fun to invite people

and have broad invitations, curations, and facilitations.

There's a lot of exciting things that people can get involved with.

So please do.

And other than that, it's been a true highlight

of the quarter and the year to have all these live streams.

And as mentioned earlier, following each live stream,

we go on to publish the proceedings

and publish them on GitHub

where they're all versioned and collaboratively edited

and also publish them to Zenodo to get the DOI and the URL.

Briefly from re-inference,

we continue work on the active inference ontology,  
active inference, robotics and embodied.

And we have several of the Institute seed projects  
related to graphical interfaces for active inference  
and cognitive agent modeling.

And then there's the plethora of individual projects  
that people are working on on their own.

Anything you want to add on any of those projects?

No, thank you.

Okay.

Well, as we head into the last months of our activity in 2020-23  
and plan to have a quiet December.

So really only two more months of activity this year  
and then we'll be in next year.

So as usual, while scrambling to prepare for the presentation  
and trying to convey what we could and should,

I developed this small meme.

So here on the bottom is the timeline, quote, timeline.

We're here in September, 2020-23.

Soon it will be classical new years  
and we'll be in 2020-24.

So here's us thinking,

wow, 2020-24 should be really exciting.

But also we remember feeling that way in 2022  
and feeling that way in 2021

but this entire concept is a nested thought from the future.

What do you think about that, Blue?

Or what are you excited about in the past, present, or future?

I mean, we don't get a treasure map, right?

So that's not something that comes with the Institute.

So it's always, I think, exciting to think about what the future holds  
and looking back, like, how we've come so far  
and, like, the leaps and what has been, like, you know,  
what has caused exponential growth and all those things.

I mean, it's always just exciting, right?

So some future thought,

maybe we'll be thinking about, you know,

how we did make those big transitions every year

and we kept making them and kept making them  
and just kept making them.

Cool.

Yeah.

one more specific call to action.

Other than the earlier communications

to actual or plausible participants, volunteers, interns,

we'd like to really make a special call

for those who are interested in joining the Scientific Advisory Board

or recommending or introducing somebody

who you think could be helpful on the SAB.

SAB is an informal body.

We have office hours with them and share documents with them.

It's very opt-in and flexible.

Other SAB members, like Chris Fields,

went on to do the course.

And just recently, Sarah Hamburg started facilitating the Morph Stream series.

So it's a way to get more involved

and have an even more structured contribution

or differently structured contribution.

And it would be amazing to have the SAB participants

as liaisons and advocates

and co-learners and co-creators

in the coming year.

Because we select the SAB on a yearly timescale,

it would be great.

It's always open to apply.

However, again, in the coming several weeks,

if we could reach out to people

who might have a positive synergistic relationship

with being on the SAB,

it would be very special and appreciated for us.

Well, that's it for this short roundtable.

So let's just give any thoughts.

Meanwhile, I will wait for anyone in the live chat

who might have questions.

If they write them in the next few minutes,

then we can address them or just talk about them.

But other than that, what's on your mind, Blue?

Or what are you excited for?

I don't know.

I'm excited for what's going to come up in 2024.

I think will be a big year for us.

I'm excited to see if we get funded or don't get funded.

But I think regardless,

we've learned a lot about where we are

and where we're going.

We've gained a lot of certainty

about our existence in the niche

and about our niche also, which is great.

Like at any time that you reduce uncertainty,

you're able to, I think,

help validate or solidify your existence

as an organization or a human or whatever

in the active inference way.

So I'm excited to kind of see how that plays out

in our future unfolding.

What about you?

Yeah, that reminds me of the bookstream series

that you facilitated with Tyler

and on Bishan Khezri's work

about kind of using active inference

to highlight the organizational generative model,

nested generative models.

And rather than talking about uncertainty

being out there in terms of the

aspects that are not known about what's out there,

it's kind of like a glass half full,

glass half empty,

starting with the certainties and uncertainties

of the generative model

as enacted through policy selection.

And so even from the very,

most informal beginnings,

like the strange loop

and the excitement was around

applying active inference  
to and by and with and for active inference  
in the multi-scale setting  
and through the bookstream,  
but also through a million other vectors,  
2023 has come through  
hugely on that.

And yet it remains a fledgling  
and nascent volunteer work  
as well.

So it's epic.

Do you have any last thoughts  
or we can conclude?

We can conclude, I think.

All right.

Thank you, Blue.

Thank you, everyone.

Hope this was a useful roundtable  
and see you next time.

Okay.

Okay, nice.

All right.

We will see you later,  
maybe tomorrow,  
maybe if I can make it.

I'm going to try.

To the discussion.

Yeah, to the discussion.

Oh, really?